

Part A · Multiple Choice

2 points each

1. **d) fine, dry, and loose, with no vegetation cover**

p. 261 — vegetation imposes a frictional drag on the wind and binds soil particles down, so an environment without it is best. Clay-rich soils resist wind erosion because clay is cohesive.

2. **a) About one-quarter**

p. 261 — in most deserts wind erosion predominates over deposition, leaving a surface of stones rather than sand.

3. **b) desert pavement**

p. 262 — also called desert lag. The interlocking stones protect the surface beneath, which is why military activity in North African deserts during World War II caused lasting degradation.

4. **b) downwind**

p. 267 — symmetrical barchans indicate wind from a constant direction. Parabolic dunes are the ones whose horns point upwind, anchored by vegetation or dampness.

5. **c) More than 50%**

pp. 263–264 — traction, or surface creep, accounts for the 20–25%. Sand generally begins moving at a wind velocity of about 4.5 m/sec.

6. **a) a playa**

p. 271 — there are more than a hundred in North American deserts, most relics of Ice Age lakes. Lake Bonneville left the Great Salt Lake, Utah Lake, and Sevier Lake behind.

Part B · True or False

1 point each

7. **TRUE**

p. 268 — dampness and plants slow the dune's base while the dry crest pushes ahead, leaving the horns anchored behind and producing the hairpin shape.

8. **FALSE**

That is a butte. Geographers use the rule of thumb that a mesa's top is wider than its height; a butte is what remains when a mesa's caprock has caved away to a small remnant. (pp. 271–273)

Part C · Short Answer

3 points each

9. **Deflation is the lifting and removal of loose fine particles by the wind; abrasion is the sandblasting of rock surfaces by sand grains carried in the wind.**

Rubric: 1 pt deflation defined, 1 pt abrasion defined, 1 pt for a valid landform for each. **Deflation** produces desert pavement (desert lag) and blowout depressions; the Qattara Depression in Egypt is partly a deflation feature.

Abrasion shapes and polishes exposed bedrock, producing ventifacts (smooth faceted rocks) and, at larger scale, yardangs — elongate ridges streamlined parallel to the prevailing wind, which archaeologists think partly form the Great Sphinx of Giza. Credit noting abrasion is limited to a meter or two above the surface, since sand grains are not lifted far. (pp. 262–263)

10. **Horizontally layered rocks are uplifted, then weathering and erosion strip the weaker layers away while a resistant caprock protects the rock beneath, leaving the resistant material standing above its surroundings.**

Rubric: 1 pt — horizontally layered rocks are uplifted by tectonic activity, and rock types differ in their ability to resist weathering and erosion. 1 pt — weaker rocks such as shale erode into valleys while resistant types (sandstone, conglomerate, quartzite, basalt, chert, limestone, lava flows, and sills) remain topographically higher; a resistant sill or lava flow often forms the flat caprock. 1 pt — the sides break into a cliff-and-bench staircase, and basal sapping erodes the soft shale beneath until the cliff collapses and retreats. When only a small remnant of caprock is left, the mesa has become a butte. (pp. 272–273)